

Smart IoT-Based Vending Machine

Using ESP32 and MQTT

Project Documentation

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Abstract

The purpose of this project is to develop a low-cost Internet of Things (IoT)-based smart vending machine that can dispense products remotely through an Android application. Unlike traditional vending machines that require cash or coin mechanisms, this system allows users to select products using a mobile application. The order is transmitted through the Internet using the MQTT communication protocol. An ESP32 microcontroller receives the order, controls the dispensing motors, monitors product delivery using infrared sensors, and reports the machine status back to the application in real time.

The proposed system demonstrates the integration of embedded systems, wireless communication, cloud services, and Android application development to create an intelligent vending solution.

1. Introduction

1.1 Background

Traditional vending machines require customers to insert coins or use payment cards before selecting a product. Such systems require dedicated payment hardware, making them expensive and difficult to maintain.

With the advancement of IoT technology, vending machines can now be connected to the Internet, allowing products to be purchased remotely through mobile applications.

The ESP32 microcontroller provides built-in Wi-Fi capability, making it suitable for cloud-connected embedded applications.

1.2 Problem Description

Traditional vending machines suffer from several limitations:

- Coin mechanisms increase hardware cost.
- Cash collection requires manual maintenance.
- Customers cannot place orders remotely.
- Machine status cannot be monitored online.
- Product inventory must be checked manually.
- Hardware failures are difficult to detect remotely.

These limitations reduce efficiency and increase maintenance costs.

1.3 Proposed Solution

- This project develops an IoT-enabled vending machine using an ESP32 microcontroller.
- Instead of using coins, users purchase products through an Android application.
- The Android application communicates with the ESP32 through the MQTT cloud server.

- The ESP32 receives dispensing commands, controls the motors, monitors the IR sensors, updates stock automatically, and reports machine status back to the cloud.

1.4 Objectives

The objectives of this project are:

- Design a low-cost smart vending machine.
- Control multiple dispensing motors.
- Detect successful product dispensing using IR sensors.
- Connect the machine to the Internet.
- Allow remote product selection through an Android application.
- Automatically update stock after dispensing.
- Detect dispensing failures.
- Publish machine status in real time.

2. Hardware Components

ESP32 Development Board

The ESP32 is the main controller of the vending machine.

Functions

- Executes the vending machine program.
- Connects to Wi-Fi.
- Connects to MQTT Cloud.
- Receives customer orders.
- Controls both motors.
- Reads IR sensors.
- Publishes machine status.

Reason for selection

- Built-in Wi-Fi
- High processing speed
- Multiple GPIO pins
- PWM motor control
- Low cost

TB6612FNG Motor Driver

The TB6612FNG controls the DC motors. The ESP32 cannot drive motors directly because its GPIO pins provide very little current. Therefore, the motor driver acts as an interface between the ESP32 and the motors.

Functions

- Motor direction control
- Motor speed control
- High current capability
- Supports two motors simultaneously

N20 12V Geared Motors

Two N20 geared motors are used.

Motor 1

- Dispenses Product A

Motor 2

- Dispenses Product B

Reason for selection

- High torque
- Small size
- Suitable for vending spirals
- Low power consumption

Infrared (IR) Sensors

Each slot contains one IR sensor.

Purpose

- Detect whether a biscuit has actually fallen.

Working principle

- When a biscuit interrupts the infrared beam, the sensor output changes from HIGH to LOW.
- The ESP32 immediately stops the corresponding motor.
- This prevents over-dispensing.

12V Power Supply

The motors require 12V operation.

The power supply provides energy to

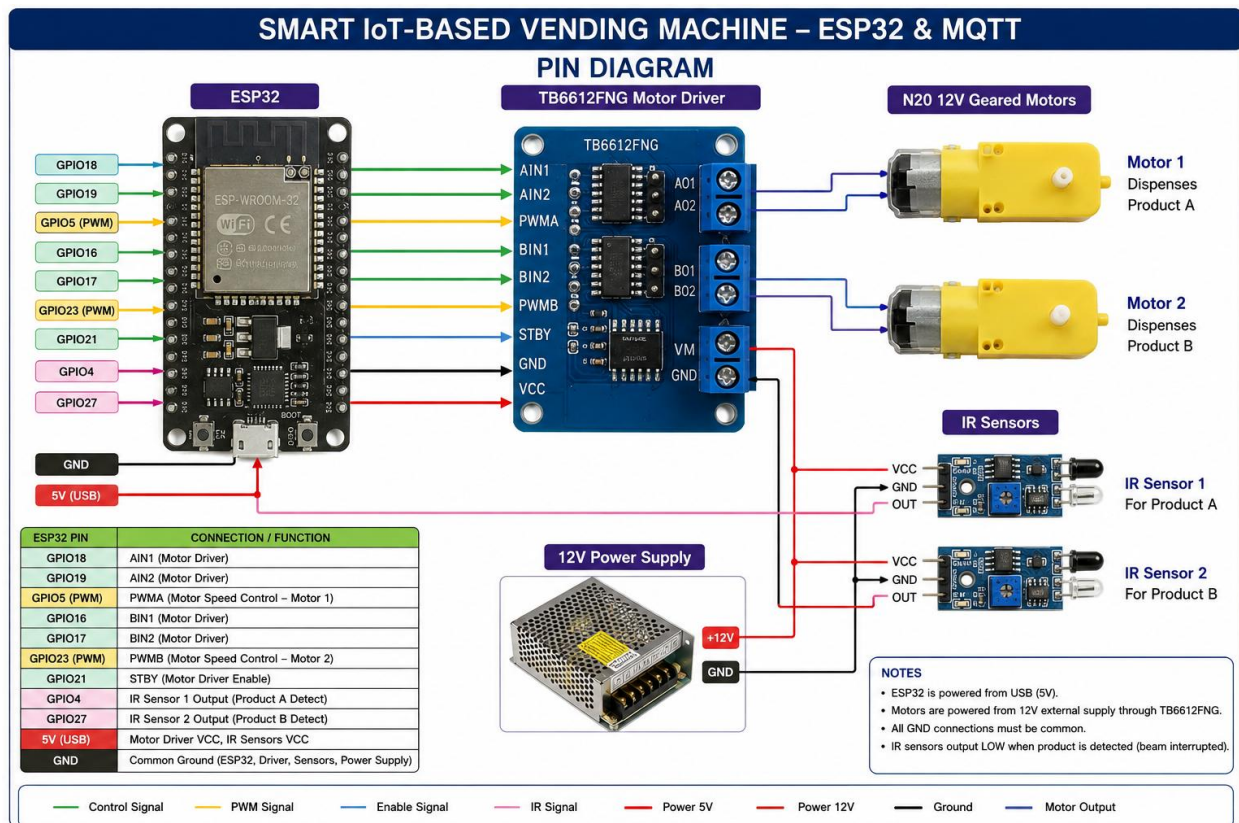
- Motor Driver
- Motors

The ESP32 is powered separately through USB or a 5V regulator.

Wi-Fi Network

The ESP32 connects to the Internet using Wi-Fi. The Wi-Fi network allows communication with the MQTT cloud server.

Pin Diagram



3. Software Components

Arduino IDE

Arduino IDE is used to write and upload the ESP32 firmware.

ESP32 Arduino Framework

Provides:

- GPIO control
- PWM generation
- Wi-Fi library

- Timer functions

PubSubClient Library

Used for MQTT communication.

Functions

- Connect to MQTT broker
- Publish messages
- Subscribe to topics
- Receive commands

ArduinoJson Library

Used for JSON parsing.

Example

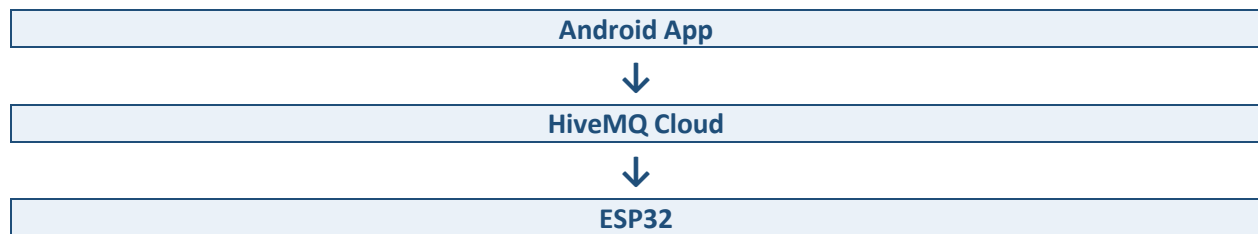
```
{
  "command": "dispense",
  "a": 2,
  "b": 1
}
```

The library converts JSON into C++ variables.

HiveMQ Cloud

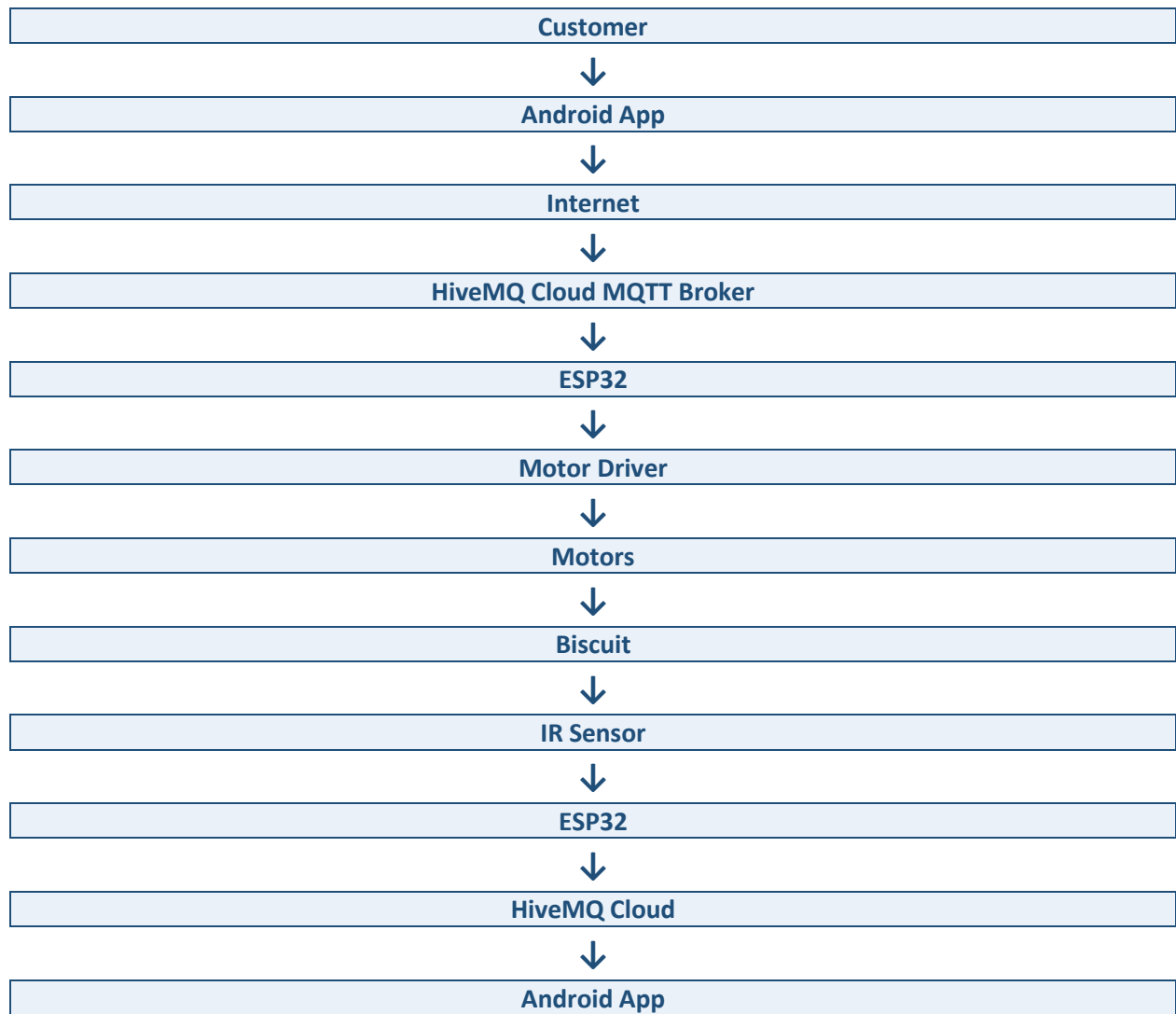
HiveMQ Cloud is the MQTT broker.

It works as the communication bridge between:



The Android application and ESP32 never communicate directly.

4. System Architecture



5. Working Procedure

Step 1

The customer opens the Android application.

Step 2

The customer selects:

- Product A
- Product B
- Quantity

Step 3

The customer presses

- Purchase

Step 4

The Android application publishes a JSON command to the MQTT broker.

Example

```
{  
  "command": "dispense",  
  "a": 2,  
  "b": 1  
}
```

Step 5

ESP32 receives the MQTT message.

Step 6

The ESP32 verifies:

- Stock availability
- Machine busy status
- Queue availability

Step 7

The order is converted into a dispensing queue.

Example

- A
- A
- B

Step 8

Motor 1 starts.

Step 9

The biscuit interrupts the IR sensor.

Step 10

- ESP32 immediately stops Motor 1.
- Stock decreases.
- Dispensed counter increases.

Step 11

The next product starts automatically.

Step 12

When every item has been dispensed, ESP32 publishes

- Order Completed

Step 13

The Android application displays

- Order Successful

6. Important Code Explanation

Wi-Fi Connection

```
WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
```

Purpose

Connect ESP32 to the wireless network.

MQTT Connection

```
mqttClient.connect(clientId.c_str(),  
MQTT_USERNAME,  
MQTT_PASSWORD);
```

Purpose

Connect ESP32 securely to HiveMQ Cloud.

MQTT Subscription

```
mqttClient.subscribe(commandTopic.c_str());
```

Purpose

Receive commands from Android.

Queue Creation

```
enqueue(1);  
enqueue(2);
```

Purpose

Store products in the order they should be dispensed.

Motor Control

```
startMotor1();
```

Starts Product A motor.

```
stopMotor1();
```

Stops Product A motor.

IR Detection

```
if(digitalRead(IR1)==LOW)
```

Purpose

Detect successful product dispensing.

Stock Update

```
qtyA--;
```

Automatically updates remaining stock.

MQTT Status Publishing

```
publishStatus(...)
```

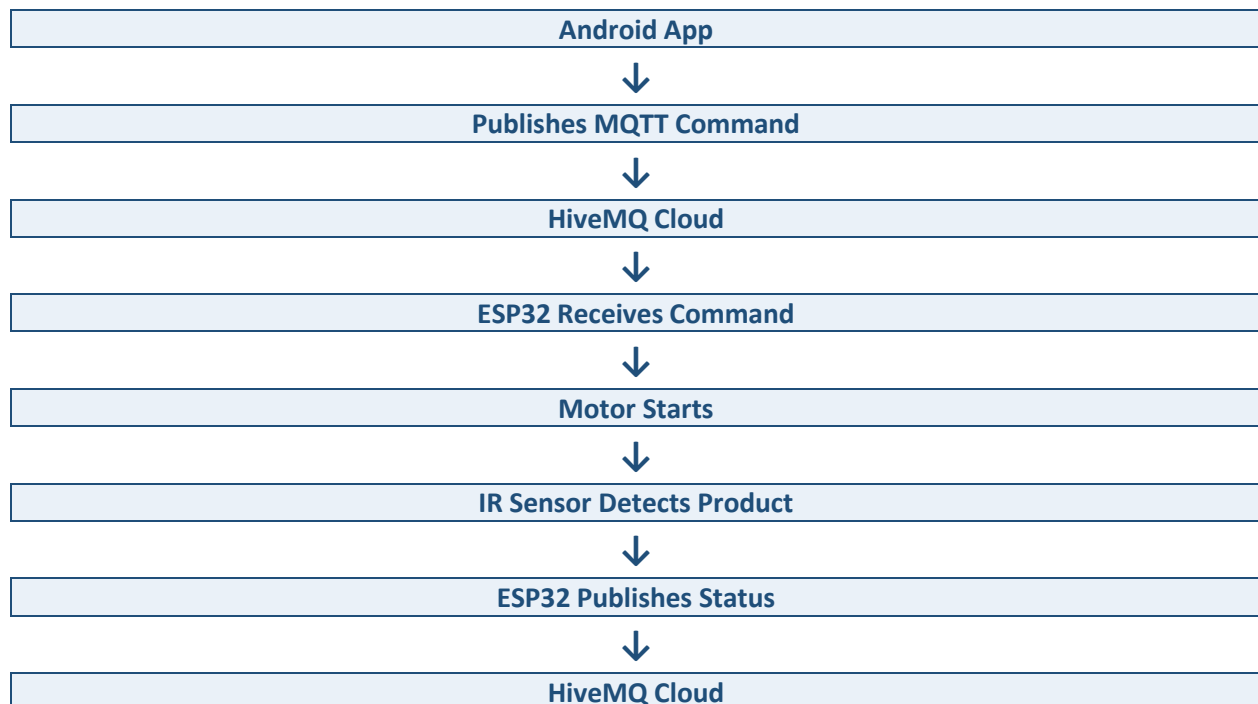
Purpose

Send machine status to the Android application.

7. Online Connectivity

This vending machine operates through the Internet using the MQTT protocol.

Communication Sequence





Android App Updates UI

Because both the Android application and ESP32 communicate through the MQTT broker, they can exchange data over the Internet as long as both devices have Internet access.

8. Advantages

- Remote operation through Internet
- Real-time machine monitoring
- Automatic stock management
- Fast communication using MQTT
- Low hardware cost
- Expandable for additional product slots
- Modular software architecture
- Fault detection through IR sensor timeout
- Secure cloud communication using TLS

9. Future Improvements

Future enhancements may include:

- Integration with real payment gateways such as bKash, Nagad, Stripe, or SSLCommerz.
- QR code-based payment.
- Product image display in the Android application.
- Database integration using Firebase.
- Remote inventory dashboard for administrators.
- Automatic low-stock notifications.
- Door lock control for maintenance.
- Camera-based monitoring.
- Sales history and analytics.
- Support for more than two dispensing slots.

10. Conclusion

This project successfully demonstrates the implementation of a smart IoT-based vending machine using the ESP32 microcontroller and MQTT communication. By replacing traditional coin-operated mechanisms with a cloud-connected Android application, the system enables remote product selection, automatic dispensing, real-time status monitoring, and inventory management. The combination of ESP32, the TB6612FNG motor driver, N20 geared motors, IR sensors, and the HiveMQ cloud broker creates a reliable, scalable, and cost-effective vending solution. The project showcases the practical application of embedded systems, IoT, wireless networking, and mobile application development, providing a strong foundation for future enhancements such as secure online payment integration, cloud databases, and advanced analytics. This demonstrates the potential of IoT technologies to modernize traditional vending systems and improve user convenience and operational efficiency.